KISII UNIVERSITY

BACHELOR OF APPLIED COMPUTER SCIENCE /SOFTWARE ENGINEERING YEAR I SEM II
COMP 102 DISCRETE MATHEMATICS

ASSIGN 2

INSTRUCTION: ***ANSWER ALL QUESTIONS***

(a) Define the power set as used in discrete mathematics (2marks)
(b) Determine the power set *P (A)* of *A* = {*a*, *b*, *c*, *d*}. (2marks)
(c) If $A = \{11,12,13\}$ and $B = \{13,14,15\}$ .Determine $A \cap B$ using a venn Diagram (4marks)
(d) If $A = \{10,11,12,13\}$ and $B = \{13,14,15\}$ show that $(A - B) \neq (B - A)$ (4marks)
(f) Let *X* = {1, 2, 3, 4}. Determine whether each relation on *X* is a function from *X* into *X*.

(*a*) *f* = {(2, 3), (1, 4), (2, 1), (3.2), (4, 4)} (2marks)

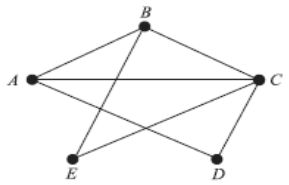
(*b*) *g* = {(3, 1), (4, 2), (1, 1)} (2marks)

(g) Two cards are drawn at random from an ordinary deck of 52 cards.

Find the probability *p* that:

(*a*) Both are spades; (*b*) one is a spade and one is a heart. (4marks)

(h) Consider the graph G in the figure below; (6marks)



Find the degree of each vertex

(i) Given *A* = {1, 2, 3, 4} and *B* = {*x*, *y*, *z*}. Let *R* be the following relation from *A* to *B*:

R = {(1, *y*), (1, *z*), (3, *y*), (4, *x*), (4, *z*)}

(*i*) Determine the matrix of the relation. (2marks)

(*ii*) Draw the arrow diagram of *R*. (2marks)

----------DUE ON 19TH DECEMBER 2025----------